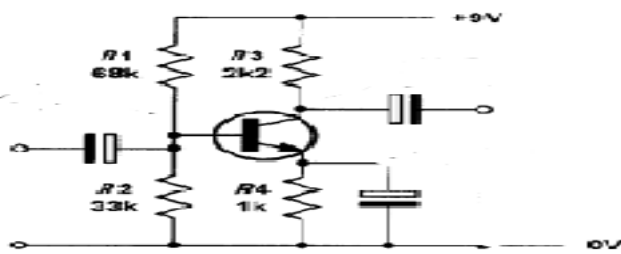


Model Question Paper
Introduction to Electronics Engineering (ESC133)

Instructions to the Candidates:

- Answer one full question from each unit
- Assume suitably any missed data

Q.No.	UNIT – I		COs	Marks
1.	a)	Explain the forward and reverse biasing conditions of PN junction diode with neat diagrams and V-I -characteristics.	CO1	(8)
	b)	Explain Light Emitting Diodes.	CO1	(6)
	c)	Explain the construction of transistors.	CO1	(6)
2	a)	A LED is to be used to indicate the presence of a 5V dc supply. If the LED has a nominal forward voltage of 2V, and is rated at a current of 12mA, determine the value of series resistor and its power rating required.	CO1	(6)
	b)	Explain BJT in common emitter configuration with the relevant characteristic curves.	CO1	(9)
	c)	A transistor operates with a collector current of 97mA and an emitter current of 98mA. Determine the value of base current and common emitter current gain.	CO1	(5)
		UNIT – II		
3.	a)	Explain the block diagram of a D.C power supply	CO2	(6)
	b)	Discuss bi-phase rectifier with reservoir circuit with relevant waveforms.	CO2	(8)
	c)	The following data was obtained during a test carried out on a d.c. power supply: Load test: Output voltage (no-load) = 12.5 V, Output voltage (600mA load current) = 12.1 V Regulation test: Output voltage (mains input, 220 V) = 12 V Output voltage (mains input, 200 V) = 11.9 V (i) Determine the equivalent output resistance of the power supply and estimate the output voltage at a load current of 400 mA. (ii) the regulation of the power supply and estimate the output voltage when the input voltage is 13.5 V .	CO2	(6)
4.	a)	Discuss bridge rectifier circuit with relevant waveforms.	CO2	(6)
	b)	In a zener voltage regulator a 10V reference voltage with a load current of 15mA is required. The available supply voltage is 25V. Select suitable components if $P_{z\max} = 400mW$.	CO2	(7)
	c)	What is a voltage multiplier? With necessary circuit discuss how the output be can tripled?	CO2	(7)

		UNIT – III		
5.	a)	Explain the classification of amplifiers.	CO3	(7)
	b)	Explain Common emitter amplifier stage with effective bias stabilization.	CO3	(7)
	c)	Explain with relevant circuit the phase shift provided by CE amplifier.	CO3	(6)
6.	a)	With frequency response curve of an amplifier explain the cutoff frequencies and bandwidth.	CO3	(5)
	b)	Determine the static value of current gain and collector voltage in the circuit shown in Fig. 6(b) if $V_B = 2V$.	CO3	(9)
				
	c)	Explain how a transistor can be used as a switch.	CO3	(6)
		UNIT – IV		
7.	a)	With the relevant implementation steps show how an exclusive OR gate can be realized using minimum number of NOR gates only.	CO4	(6)
	b)	Realize full adder using two half adders.	CO4	(8)
	c)	Differentiate between Latch and Flip Flop.	CO4	(6)
8.	a)	Simplify and hence realize the following expression using basic gates (i) $ABC + \bar{A}B + \bar{B}C$ (ii) $(\bar{A} + C)(\bar{B} + C)(A + \bar{B})$	CO4	(7)
	b)	Explain the functionality of J K Flip Flop along with its applications.	CO4	(7)
	c)	Realize Full adder using logic gates.	CO4	(6)
		UNIT – V		
9.	a)	Draw & explain the block diagram of microprocessor based system.	CO5	(10)
	b)	Explain the operation of microprocessor system.	CO5	(10)
10.	a)	Draw & explain the internal architecture of 8 bit microprocessor.	CO5	(10)
	b)	Draw & explain the block diagram of microcontroller based system.	CO5	(10)

